Curriculum Vitae – Luca Garau

Personal Information

Luca Garau
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I am a national PhD student in AI & Society at ISTI-CNR and the University of Pisa, in collaboration with the University of Cagliari. I mainly work in the field of Digital Fabrication and Geometry Processing.

Current Position

National PhD in AI & Society | **University of Pisa | ISTI-CNR** | **2026 – Present**
National PhD in AI & Society at ISTI-CNR and the University of Pisa, in collaboration with the University of Cagliari. The research work focuses on Digital Fabrication and Geometry Processing, particularly in the computational design of multi-piece rigid molds.
Supervisors: Dt. Daniela Giorgi, Prof. Gianmarco Cherchi, Dt. Luigi Malamo

Education

Master's Degree in Computer Science (Graphics and Vision) | **University of Cagliari** | **2023 – 2025**
Grade: 100/100 cum laude
Thesis: *"Classification and Robust Management of Triangle Intersections Using Alternative Numerical Representations"*
Thesis focused on the development of a novel approach for detecting and classifying triangle intersections independently of the underlying numerical representation system.
Supervisor: Dott. Gianmarco Cherchi

Bachelor's Degree in Computer Science | **University of Cagliari** | **2021 – 2023**
Grade: 110/110 cum laude
Thesis: *"An AI Approach to EEG-based Emotion Recognition using a Consumer Device"*
Recognition of emotions via EEG signals using a Machine Learning approach.
Supervisor: Prof. Daniele Riboni

SGP 2026 Phd School | **SGP Conference - Universitat Bern** | **2026**
PhD School of SGP 2026 conference.

New Mathematical Methods in Geometric Processing | **University of Bonn** | **2026**
PhD School on New Mathematical Methods in Geometric Processing.

EG PhD School | **Eurographics Italian Chapter** | **2025**
PhD School on Geometric Intelligence: Theory and Practice of 3D Understanding in Computer Graphics and 3D Computer Vision.

EG PhD School | **Eurographics Italian Chapter** | **2024**
PhD School on Modern Computer Graphics: AI Techniques for 3D Reconstruction, Rendering, and Analysis.

Samsung Innovation Campus | **Samsung Electronics Italia S.p.A.** | **2021**
Internet of Things course in collaboration with the University of Cagliari (top grade, 30L).

Experience

Research Fellow | **University of Cagliari — DMI** | **2025 – 2026**
Research grant with the Department of Mathematics and Computer Science (DMI) of the University of Cagliari in Computer Graphics and Geometry Processing.

Research Fellow | **University of Cagliari — DIEE** | **2024 – 2025**
Research grant with the Department of Electrical and Electronic Engineering (DIEE) of the University of Cagliari on the analysis of mesh coding configurations for adaptive streaming of immersive contents in virtual reality environments.

Teaching Assistant — Databases | **University of Cagliari** | **2024 – 2026**
Assistance with laboratory lessons aimed at teaching the SQL language.

Teaching Assistant — Numerical Methods | **University of Cagliari** | **2023 – 2025**
Lecture support and lab assistance for the course of Scientific Calculus and Numerical Methods. The course covers fundamentals of numerical analysis, linear algebra, function approximation, nonlinear equation solving, and includes practical exercises in Matlab.

Publications

Luca Garau, Gianmarco Cherchi. A robust approach to detect intersections between triangles with different numerical representations. International Conference in Central Europe on Computer Graphics, Visualization and Computer Vision (WSCG) | **2025**
Full paper presented at WSCG 2025. Introduces a novel and robust approach for detecting and classifying triangle intersections, along with an open-source C++ header-only tool.
DOI: 10.24132/CSRN.2025-7

Luca Garau, Riccardo Scateni, Gianmarco Cherchi. Triangle-triangle intersections with different numerical representations. IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW), pp. 401–404 | **2025**
Short paper presented at the ReDigiTS 2025 workshop at IEEE VR 2025: a work-in-progress on a robust intersection-detection tool independent of the numerical representation used.
DOI: 10.1109/VRW66409.2025.00092

Conferences Attended

SGP 2026 | **2026**
Symposium on Geometry Processing, 1–3 July 2026, Bern, Switzerland

STAG 2025 | **2025**
Smart Tools and Applications in Graphics, 27-28 November 2025, Genoa, Italy

WSCG | **2025**
International Conferences in Central Europe on Computer Graphics, Visualization and Computer Vision, 26–29 May 2025, Pilsen, Czech Republic
Presented: *A robust approach to detect intersections between triangles with different numerical representations*

IEEE VR | **2025**
The 32nd IEEE Conference on Virtual Reality and 3D User Interfaces, 08–12 March 2025, Saint-Malo, France
Presented: *Triangle-triangle intersections with different numerical representations*

STAG 2024 | **2024**
Smart Tools and Applications in Graphics, 14–15 November 2024, Verona, Italy

EICS | **2024**
The 16th ACM SIGCHI Symposium on Engineering Interactive Computing Systems, 24–28 June 2024, Cagliari, Italy

Skills

Programming languages: Python, C, C++, C#, Java, Matlab, SQL.
Operating systems: Windows, Unix/Linux, macOS.
Languages: Italian (mother tongue), English (fluent).